

OLAP Cube Description

A/L Student Results Dataset — AL.csv

1. Introduction

This report describes an Online Analytical Processing (OLAP) cube designed using the A/L student results dataset. The purpose of the cube is to support multidimensional analysis of student participation and academic performance across different categories.

The source dataset contains 337,553 student/result records. The cube is designed to answer questions such as how many students belong to each stream, how performance differs by gender and syllabus, and how results can be examined at progressively more detailed levels.

2. Business Requirements

Requirement	Analysis supported
Student participation	Count students by stream, gender and syllabus.
Performance comparison	Compare average Z-scores between groups.
Syllabus comparison	Compare New and Old syllabus results.
Gender analysis	Analyze performance and participation by gender.
Subject analysis	Drill down from stream to subject and grade.

3. Data Source

The data source is the file **AL.csv**. The dataset contains A/L student result information including stream, gender, syllabus, subjects, grades, Z-score, district rank and island rank. The available A/L year is 2020 only.

4. OLAP Cube Design

The central cube uses three primary dimensions: **Stream × Gender × Syllabus**. Each cell represents an observed combination of these dimensions and contains aggregated performance measures.

Component	Definition
Fact	A/L student/result record
Dimension 1	Stream
Dimension 2	Gender
Dimension 3	Syllabus (New/Old)
Drill-down	Stream → Subject → Grade
Time	A/L Year (2020 only)

5. Measures

Measure	Type	Use
Student Count	Additive	Can be summed across dimensions.
Average Z-score	Non-additive	Must be recalculated when aggregation level changes.
Average District Rank	Non-additive	Ranks should not be summed.
Average Island Rank	Non-additive	Ranks should not be summed.

6. OLAP Operations

Roll-up

Roll-up summarizes detailed cube data at a higher level. For example, the cube can move from Stream × Gender × Syllabus to a Stream-level summary. Student Count is summed, while average measures are recalculated.

Slice

A slice fixes one dimension to a single value. In this implementation, a female-only slice selects Gender = FEMALE and summarizes the remaining dimensions.

Dice

A dice operation selects multiple values from multiple dimensions. The example selects ARTS and COMMERCE streams together with NEW and OLD syllabus categories.

Drill-down

Drill-down moves from a high-level summary to more detailed information. The selected hierarchy is Stream → Subject → Grade, allowing an analyst to examine subject-level grade distributions within a stream.

7. Conceptual Cube

Stream × Gender × Syllabus

Each cube cell contains:

- Student Count
- Average Z-score
- Average District Rank
- Average Island Rank

8. Validation

Validation item	Expected result
Source records	337,553
Cube student total	337,553
Record reconciliation	PASS

Validation item	Expected result
Stream dimension	Validated
Gender dimension	Validated
Syllabus dimension	Validated
Roll-up	Validated
Slice	Validated
Dice	Validated

9. Conclusion

The resulting OLAP cube provides a structured way to analyze the A/L dataset from multiple perspectives. Stream, gender and syllabus form the main dimensions, while student count and performance statistics form the measures. Roll-up, slice, dice and drill-down operations allow users to move between summary and detailed views.

Important limitation: the dataset contains only the 2020 A/L year. Therefore, the current cube cannot support meaningful year-over-year trend analysis. Additional years could be added later as a richer Time dimension.